GOODREADS BOOK REVIEW

DATA SOURCES

source:
https://www.kaggle.com/datasets/jealousleopard/goodreadsbooks

kaggle

Format: csv
License CC0: Public Domain
5-Star Open Data: ★★★

WIKIPEDIA

Format: HTML
License: GNU
5-Star Open Data: ★★

Overview

This project focuses on transforming structured data from the Goodreads Books Dataset into a semantic knowledge graph using RDF and OWL technologies. The primary goal is to enable rich querying, semantic reasoning, and visual exploration of book-related data such as titles, ratings, reviews, authors, and publication details.

Ontology



Prefixes

```
@prefix gr: <https://gitlab.dit.htwk-leipzig.de/
             minh_huy.nguyen/goodread-book-review>
@prefix xsd: <http://www.w3.org/2001/XMLSchema#>
@prefix owl: <http://www.w3.org/2002/07/owl#>
@prefix rdfs:<http://www.w3.org/2000/01/rdf-schema#>
```

PROCESS

1. CSV Parsing: Parse the Goodreads CSV and map fields to ontology terms.
2. Ontology Modeling: Define classes and relationships in OWL to structure the data.
3. RDF Generation: Generate .ttl files from parsed data.
4. SPARQL Integration: Load data and query using SPARQL.
5. Frontend: Visualize results and enable dynamic query filtering.

I-ITWK
Hochschule für Technik, Wirtschaft und Kultur Leipzig

Semantic Web
Sommer-semester 25

22INB

Anastasia Christabel Utomo
Minh Huy Nguyen
Coffi Gracia Wilfrid Akplogan